■ Supplementary document

Hi, why does a block of iron sink while a block of wood floats? Why is a gold ring so heavy for its size? More than 2,000 years ago, King Archimedes wondered if his crown was pure gold. How do you know without melting it? The answer lies in two words, density. In this video, you will learn how to calculate a density, recognize a material using it, and understand alloys and composite materials. Part one, metals. The vast majority of known chemical elements are metals, iron, copper, aluminum, nickel, gold and many others. In nature, there are very few pure metals. Most are enclosed in ores, rocks from which the industry extracts the metal. New Caledonia has one of the largest nickel reserves in the world. The ore is extracted in the mines and then processed in the factory to obtain the metal. Part two, density. The density, note rho, is the mass divided by the volume, rho equals m divided by v. It is expressed in kilograms per cubic meter or in grams per cubic centimeter. One gram per cubic centimetre is 1,000 kilograms per cubic metre. Water is one gram per cubic centimetre and air about 1.2 kilograms per cubic metre. Example, an aluminum cube 2 centimeters wide with a volume of 8 cubic centimeters and a mass of 21.6 grams. Its density, 21.6 divided by 8, is 2.7 grams per cubic centimeter. Each material has its own density. This is a characteristic property. Aluminum, 2.7. Iron, 7.9. Copper, 9. Lead, 11.3. However, 19.3. It makes it possible to recognize a material. An object of 540 grams per 200 cubic centimeters. 540 divided by 2 cents, 2.7. It's aluminum. It also predicts buoyancy. A solid object floats in water if its density is less than one gram per cubic centimeter. Wood floats, iron sinks. Third part, metals and alloys. An alloy is a mixture of a metal with other elements. It is often stronger than pure metal. Steel, for example, is iron with a little carbon. When the added atoms are smaller, they slip between the atoms of the metal. This is insertion. In steel, small carbon atoms are inserted between the iron atoms. When the atoms added are close in size, they take the place of certain atoms of the metal. It's substitution. Brass is copper, some atoms of which are replaced by zinc. Every alloy has its properties. Stainless steel, made of iron, chromium and nickel, does not rust. Much of New Caledonia's nickel is used to make it. Part Four, Composite Materials. A composite material combines at least two immiscible materials. Reinforced concrete, for example, combines concrete and steel bars. In reinforced concrete, concrete resists crushing and steel resists stretching. Together, they do better than each one alone. The properties of a composite depend on the assembly. Carbon fiber and resin give light and strong parts, used in bicycles, airplanes or surfboards. So, what's the point? All this, not just to pass a test. Choosing a material means choosing its properties. Lightweight for an airplane, strong for a deck, stainless for a sink. And with the density, we can verify that an ingot is indeed gold, without melting it as in the story of Archimedes. These concepts are used in industry, construction, jewellery and research. To summarize, density is worth mass divided by volume and characterizes a material. An alloy is formed by insertion or substitution. A composite assembles immiscible materials. It's up to you to play, a 135 gram object occupies 50 cm3. What is its density? Pause the video. 2.7 grams per cm3, 135 divided by 50. This is the density of aluminum. The next time you lift an object, ask yourself if it is light or heavy for its size. Subscribe for the rest and see you soon.